

A dark blue vertical bar is positioned to the left of the word "TRANSISTORS".

# TRANSISTORS

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# TRANSISTORS:

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- ▶ Transistor is a miniature (small in size) device used to control and regulate the flow of electric signals
- ▶ A type of semi conductor device that is used to conduct or insulate electric current (I) and voltage (V).
- ▶ A transistor can work as:
  - ▶ Switch (ON/OFF), and
  - ▶ Amplifier.



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# TRANSISTORS

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graph TD; TRANSISTORS --> BJT; TRANSISTORS --> FET; BJT --> PNP; BJT --> NPN; FET --> JFET; FET --> MOSFET;
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BJT

FET

PNP

NPN

JFET

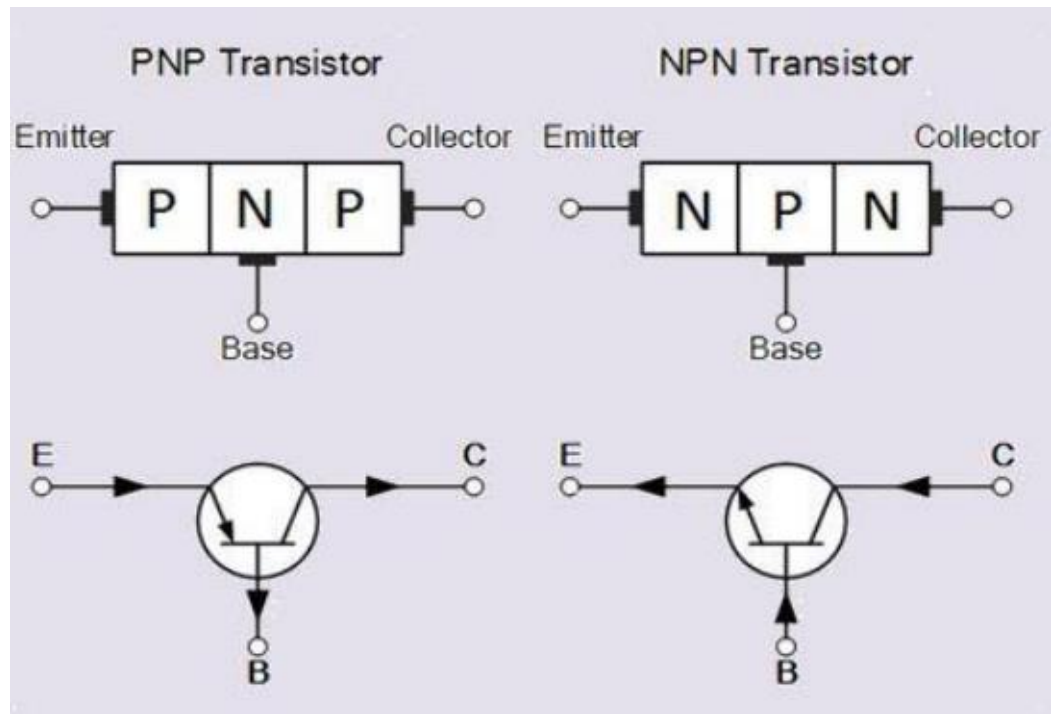
MOSFET

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# BASIC FEATURES OF BJT:

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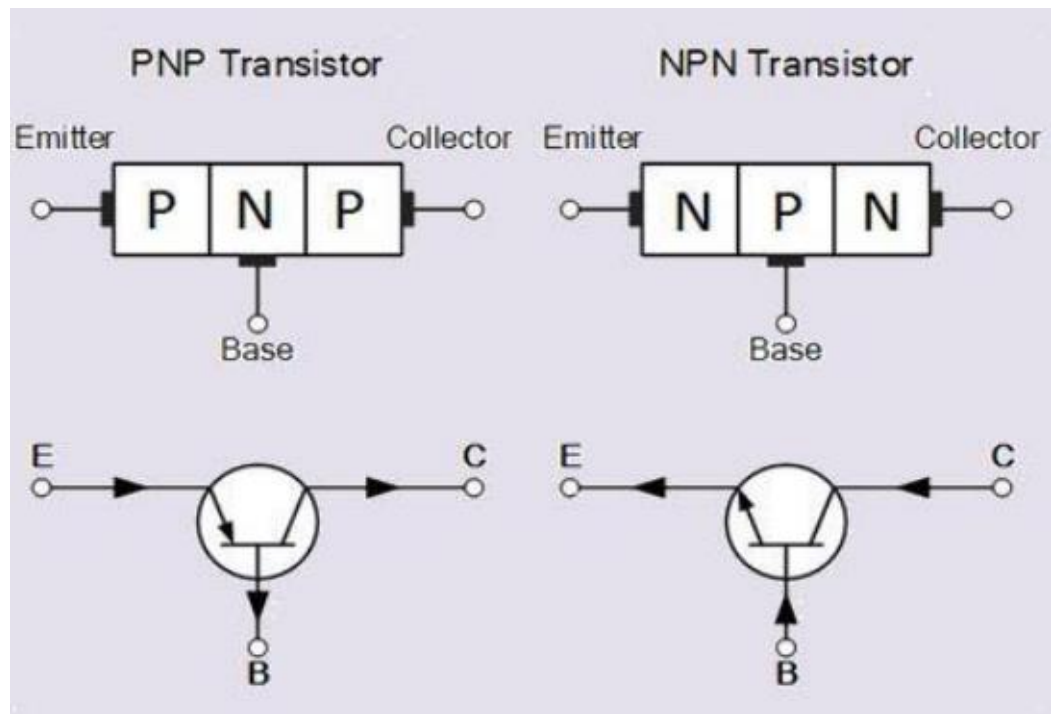
- ▶ Three region exist in BJT: Base, Emitter, and Collector,
- ▶ Base is in between emitter and collector,
- ▶ Base is thin and collector is wide.



# PN JUNCTION IN BJT:

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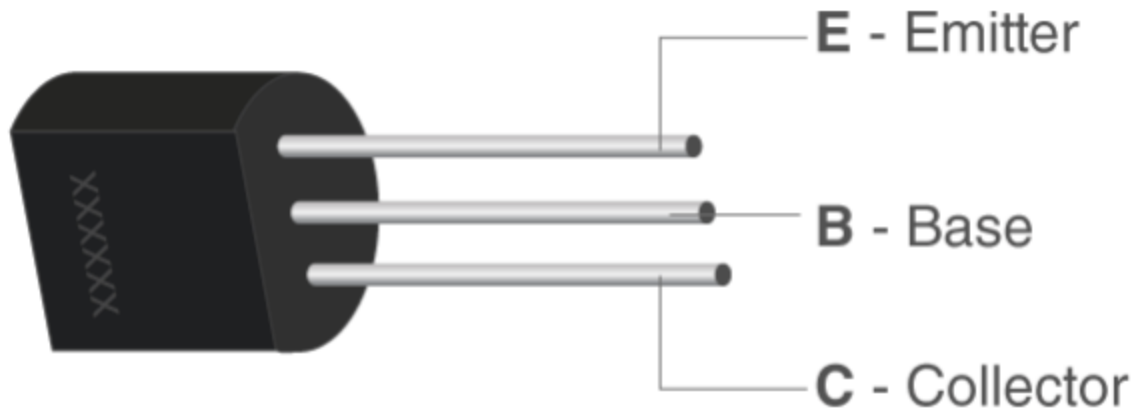
- ▶ Junction 1: Between emitter and base,
- ▶ Junction 2: Between base and collector.



# COMPONENTS OF BJT:

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- ▶ **Emitter:** It is the side that collects carriers
- ▶ **Base:** This is used to activate the transistor.
- ▶ **Collector:** It is the side that supplies carriers .

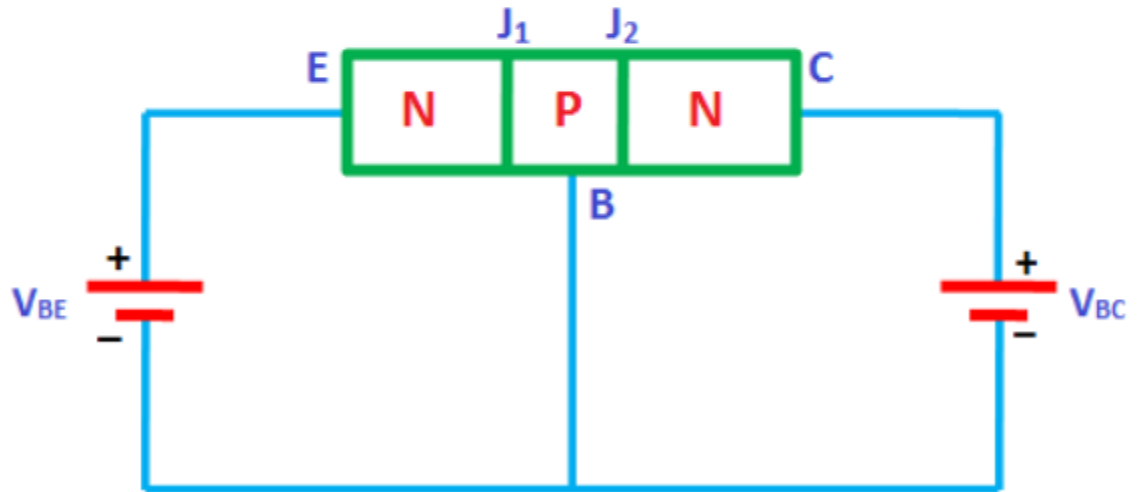
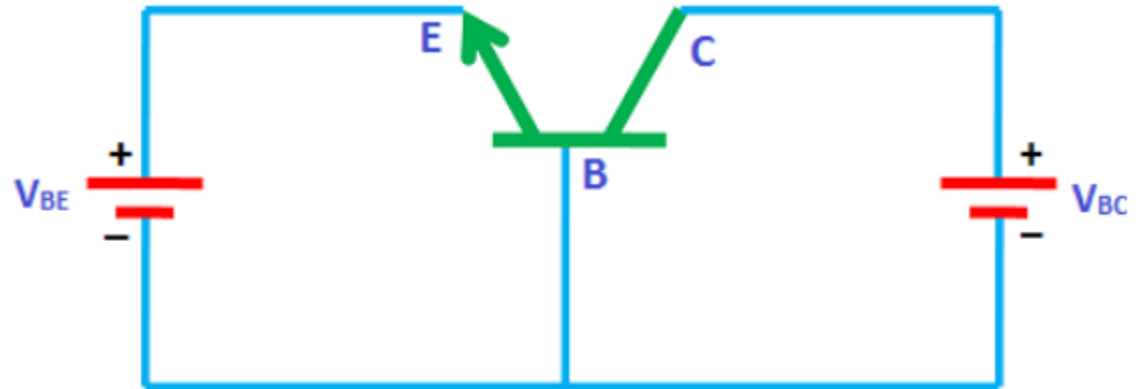


| <b><i>EMITTER</i></b>                                                                                                                                                                                                                                                                                                                                          | <b><i>BASE</i></b>                                                                                                                                | <b><i>COLLECTOR</i></b>                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>i. The section of one side that supplies carriers is called emitter.</li> <li>ii. Emitter is always forward biased w.r.t base so it can supply carrier.</li> <li>iii. For “npn transistor” emitter supply electrons to its junction.</li> <li>iv. For “pnpt ransistor” emitter supply holes to its junction.</li> </ul> | <ul style="list-style-type: none"> <li>i. The middle section which forms two pn junction between emitter and collector is called Base.</li> </ul> | <ul style="list-style-type: none"> <li>i. The section on the other side that collects carrier is called collector.</li> <li>ii. The collector is always reversed biased w.r.t base.</li> <li>iii. For “npn transistor” collector receives electrons to its junction.</li> <li>iv. For “pnp transistor” collector receives holes to its junction.</li> </ul> |



# BJT MODES OF OPERATION:

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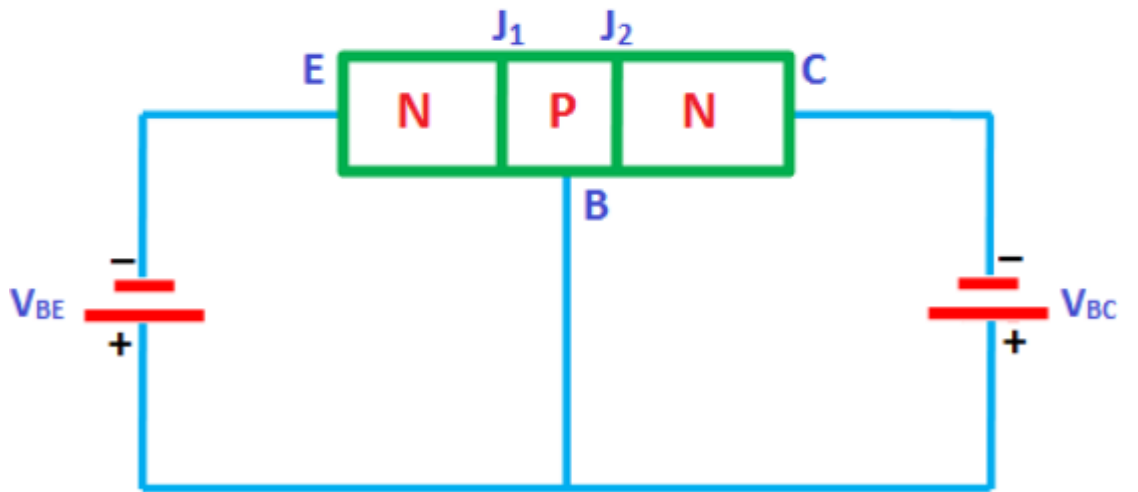
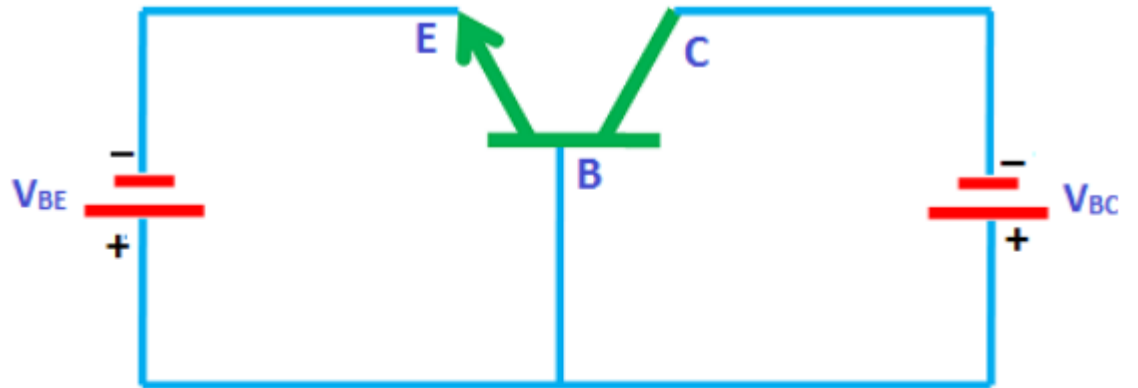
Cutoff mode





# BJT MODES OF OPERATION:

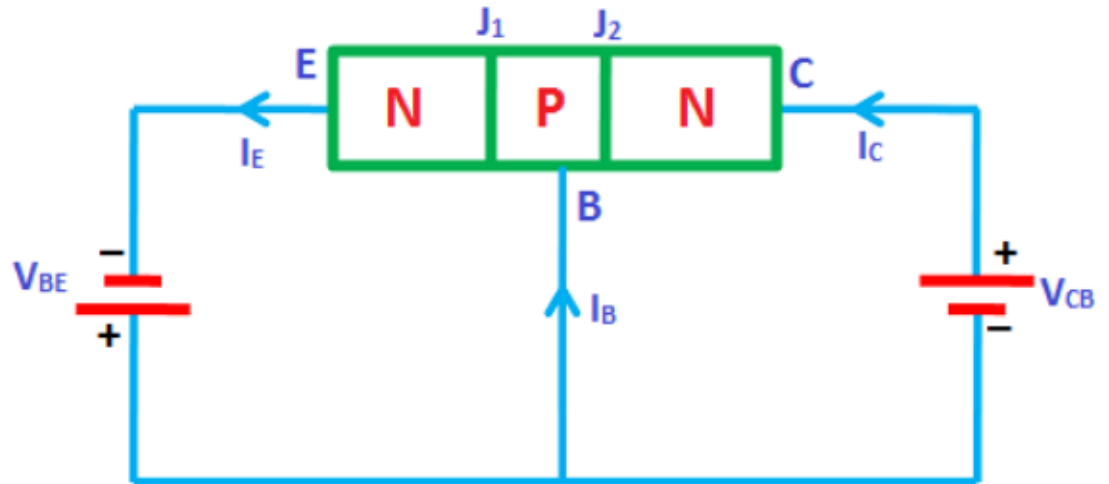
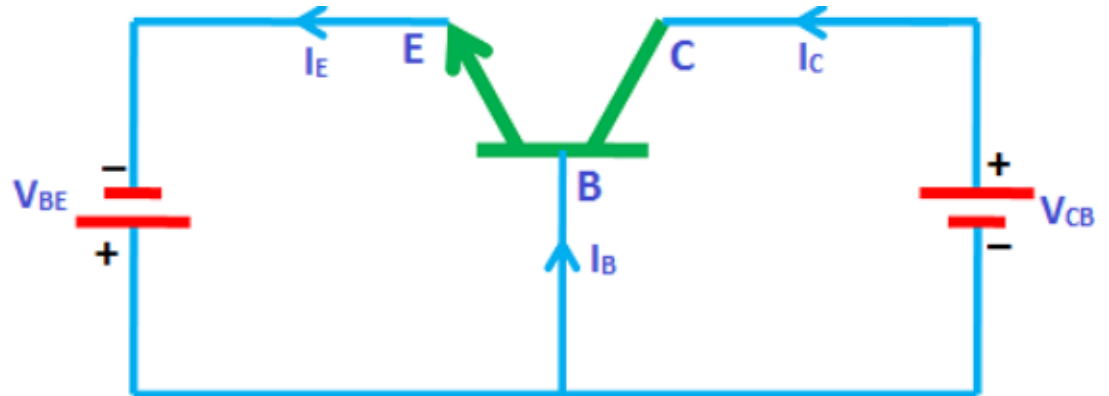
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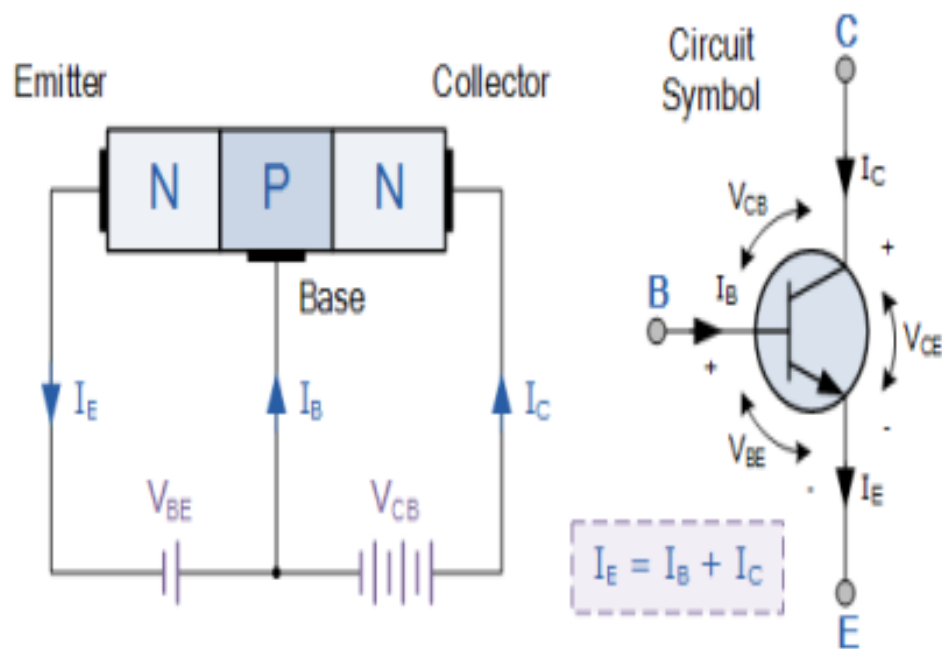
Saturation mode



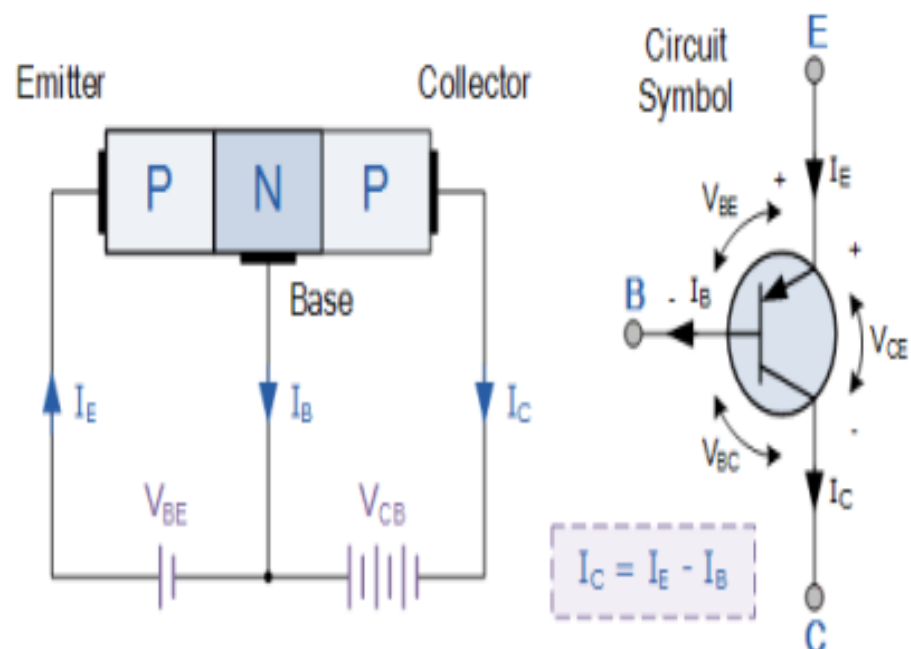
# BJT MODES OF OPERATION:



Active mode



**NPN TRANSISTOR**



**PNP TRANSISTOR**

